

5 Gas exchange requires diffusion.

The rate of diffusion is dependent on the surface area, the diffusion distance and the concentration gradient.

The rate of diffusion can be calculated using the formula:

$$\text{rate of diffusion} \propto \frac{\text{surface area} \times \text{concentration gradient}}{\text{diffusion distance}}$$

(a) Name the law on which this formula is based.

(1)

(b) What will happen to the rate of diffusion if all three of the following are changed at the same time?

- the surface area is halved
- the concentration gradient is halved
- the diffusion distance is halved

(1)

- ☐ **A** it will stay the same
- ☐ **B** it will be a quarter of its value
- ☐ **C** it will halve
- ☐ **D** it will double

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(c) The properties of gas exchange surfaces include:

- a large surface area
- a short diffusion distance
- a high concentration gradient.

Describe how mammalian lungs are adapted to have each of these three properties.

(3)

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*(d) The photograph shows a hellbender salamander.

skin folds



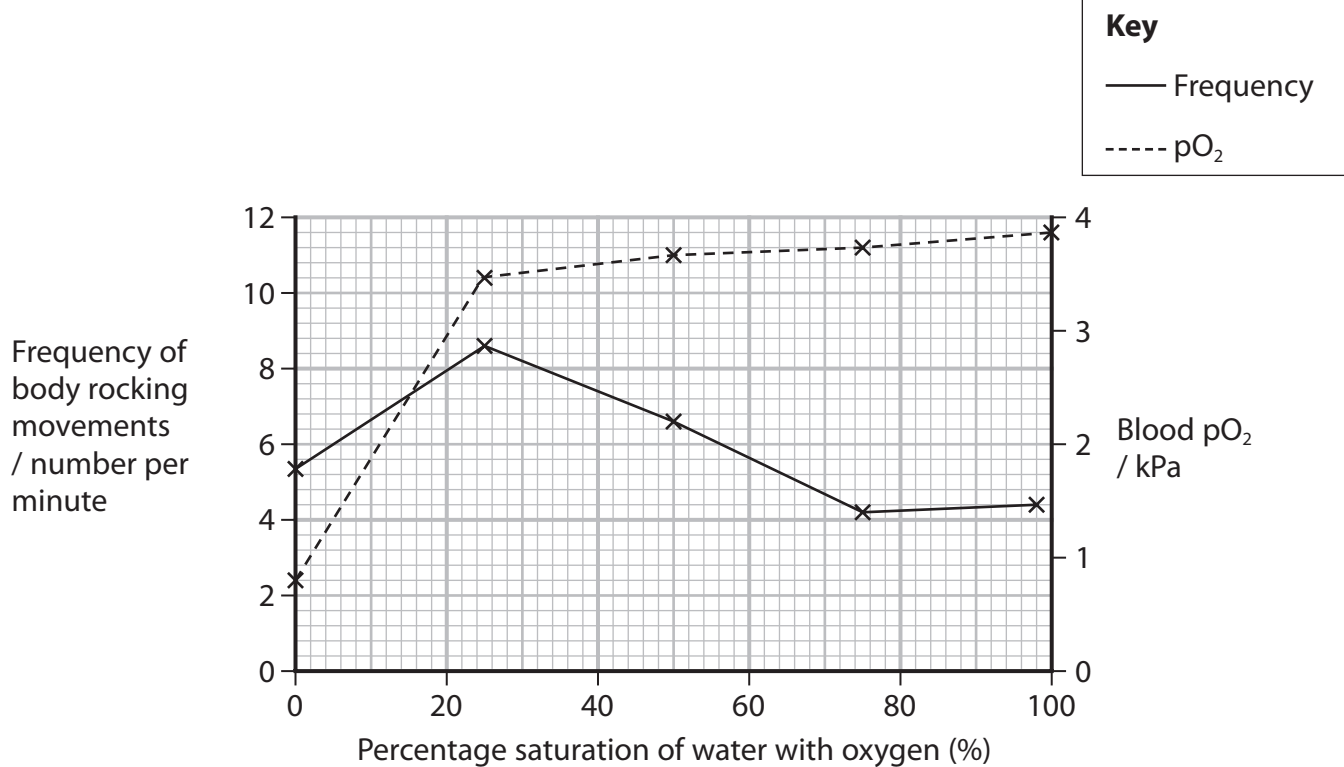
(Source: © Robert Hamilton/Alamy Stock Photo)

These salamanders live in freshwater streams that are shallow and fast-flowing.

They use their lungs and the surface of their skin for gas exchange.

They also use body rocking and swaying movements in the water.

The graph shows the relationship between the frequency of body rocking movements, blood pO_2 and the percentage saturation with oxygen of the water they live in.



Discuss how the structure and behaviour of these salamanders are adapted for gas exchange.

(6)

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(Total for Question 5 = 11 marks)

